



NORTH SYDNEY BOYS HIGH SCHOOL

2013 YEAR 12 ASSESSMENT TASK 1

Mathematics Extension 1

General Instructions

- **Working time – 60 minutes**
- **Reading time – 5 minutes**
- Answer all questions on the lined paper in the booklet provided.
- Write using blue or black pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question.
- Each new question is to be started on a **new page** with the exception of the multiple choice which can be answered on the same page.
- Attempt all questions.

Class Teacher: Please colour

- ☐ Mr Berry
- ☐ Mr Fletcher
- ☐ Mr Lam
- ☐ Mr Lin
- ☐ Mr Lucas
- ☐ Ms Ziazaris

Student

No./Name: _____

(To be used by the exam markers only.)

Question No	1-4	5	6	7	8	9	10	Total	%
Mark	$\frac{4}{4}$	$\frac{8}{8}$	$\frac{12}{12}$	$\frac{3}{3}$	$\frac{8}{8}$	$\frac{7}{7}$	$\frac{8}{8}$	$\frac{50}{50}$	$\frac{100}{100}$

ANSWER ALL QUESTIONS IN THE ANSWER BOOKLET PROVIDED

QUESTION 1

A parabola has equation $(x + 3)^2 = -16y$.

The coordinates of the focus will be

- A. $(-3, 4)$
- B. $(-3, -4)$
- C. $(3, 4)$
- D. $(3, -4)$

QUESTION 2

Find the general solution of $2 \cos x = \sqrt{3}$.

- A. $2\pi n \pm \frac{\pi}{6}$
- B. $\pi n \pm \frac{\pi}{6}$
- C. $2\pi n + \frac{\pi}{6}$
- D. $\pi n + \frac{\pi}{6}$

QUESTION 3

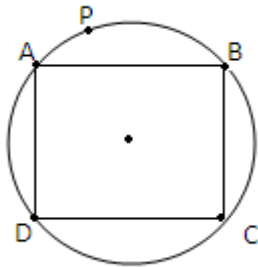
How many points of intersection does the line $y = mx - 2m^2$ have with the parabola $x^2 = 8y$.

- A. Unable to distinguish
- B. 0
- C. 1
- D. 2

exam continues overleaf.....

QUESTION 4

ABCD is a square inscribed in a circle. P is a point on the minor arc AB. Find the size of $\angle APB$.



A. 90

B. 135

C. 45

D. 120

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QUESTION 5

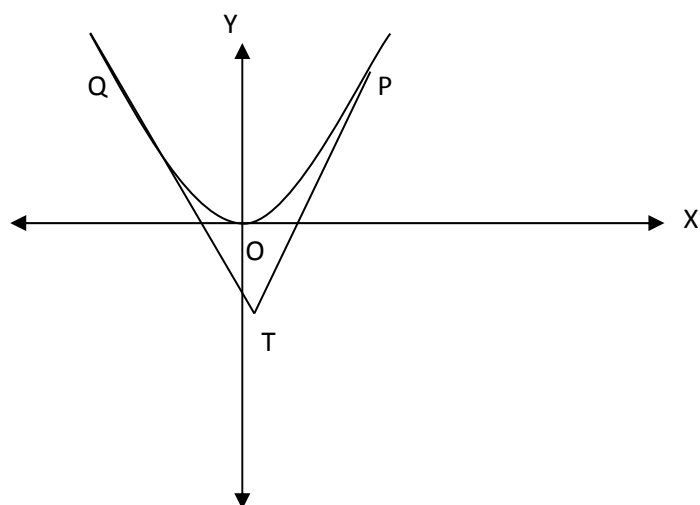
- (a) Find the Cartesian equation of the curve whose parametric equations are 2

$$x = 2 \cos \theta$$

$$y = \sqrt{3} \sin \theta$$

- (b) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$.
The tangents at P and Q meet at T .

- (i) Derive the equation of the tangent at P . 2
(ii) Show that the coordinates of T are $[a(p+q), apq]$. 3
(iii) If T lies on the line $y = -2a$ find a relationship between p and q . 1



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QUESTION 6

(a) (i) Prove that $\frac{\sin 3x}{\cos 2x \cos x} = \tan 2x + \tan x$ 2

(ii) Hence, solve $\tan 2x + \tan x = 0$ for $0 \leq x \leq 2\pi$ 2

(b) Given that $\tan \alpha = -\frac{2}{3}$ and $\sin \beta = \frac{1}{4}$ and $\frac{\pi}{2} < \alpha < \beta < \pi$.

Find the exact value for

(i) $\sin 2\alpha$ 2

(ii) $\cos(\beta - \alpha)$ 3

(c) The lines $y = mx$ and $y = 2mx$, where $m > 0$ are inclined to each other at 3

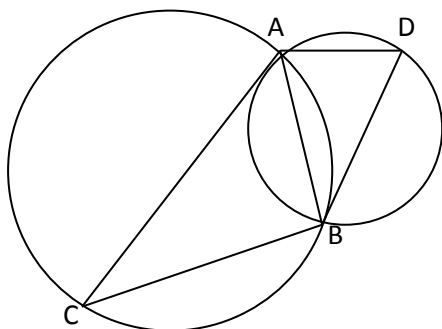
an angle θ such that $\tan \theta = \frac{1}{3}$. Find the possible values of m .

QUESTION 7

AC is a tangent to circle ABD and BD is a tangent to circle ABC. AB is a common chord, meeting the tangents at A and B.

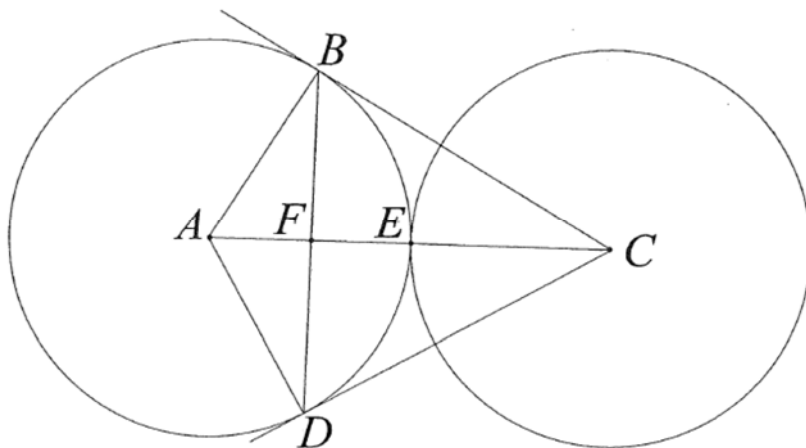
(i) Copy or trace diagram into your booklet

(ii) Prove $AD \parallel CB$ 3



exam continues overleaf.....

Question 8



Two circles with equal radii and centres A and C touch externally at E as shown in the diagram.

The lines BC and DC are tangents from C to the circle with centre A.

- | | | |
|-------|--|---|
| (i) | Copy or trace the diagram into your booklet. | |
| (ii) | Explain why ABCD is a cyclic quadrilateral. | 2 |
| (iii) | Show that E is the centre of the circle that passes through A, B, C and D. | 2 |
| (iv) | Show that $\angle BCA = \angle DCA = 30^\circ$. | 2 |
| (v) | Deduce that $\triangle BCD$ is equilateral. | 2 |

exam continues overleaf.....

QUESTION 9

- (a) Consider the quadratic function $x^2 - (k + 2)x + 4 = 0$. For what values of k is the quadratic function positive definite. 2
- (b) The quadratic equation $ax^2 + bx + c = 0$ has roots $x = \tan \alpha$ and $x = \tan \beta$.
- (i) Show that $\tan(\alpha + \beta) = \frac{b}{c - a}$ 2
- (ii) Show that $\tan^2(\alpha - \beta) = \frac{b^2 - 4ac}{(a + c)^2}$ 3

Question 10

The straight line $y = mx + b$ meets the parabola $x^2 = 4ay$ at the points

$P(2ap, ap^2)$ and $Q(2aq, aq^2)$.

- (a) Derive the equation of the chord PQ and hence or otherwise, show that $pq = -\frac{b}{a}$. 2
- (b) Prove that $p^2 + q^2 = 4m^2 + \frac{2b}{a}$ 2
- (c) Given that the equation of the normal to the parabola at P is $x + py = ap^3 + 2ap$, and that, N, the point of intersection of the normals at P and Q has the coordinates $[-apq(p + q), a(2 + p^2 + pq + q^2)]$, express these coordinates in the terms of a, m and b. 2
- (d) Suppose that the chord PQ is free to move while maintaining a fixed gradient. Find the locus of N, and show that it is a normal to the parabola. 2

END OF EXAMINATION

Suggested Solutions

Section I

(Lin)

1. (B) 2. (A) 3. (C) 4. (B)

Section II

Question 5 (Berry)

(a) (2 marks)

$$\begin{cases} x = 2 \cos \theta \\ y = \sqrt{3} \sin \theta \end{cases}$$

Make $\cos \theta$ and $\sin \theta$ the subject,

$$\begin{cases} \frac{x}{2} = \cos \theta \\ \frac{y}{\sqrt{3}} = \sin \theta \end{cases}$$

Squaring both equations and adding,

$$\begin{aligned} \left(\frac{x}{2}\right)^2 + \left(\frac{y}{\sqrt{3}}\right)^2 &= \cos^2 \theta + \sin^2 \theta \\ \frac{x^2}{4} + \frac{y^2}{3} &= 1 \end{aligned}$$

(b) i. (2 marks)

$$\begin{aligned} x^2 &= 4ay \\ y &= \frac{1}{4a}x^2 \\ \frac{dy}{dx} &= \frac{2}{4a}x = \frac{1}{2a}x \Big|_{x=2ap} \\ &= \frac{2ap}{2a} = p \end{aligned}$$

$\therefore$ gradient of tangent at $P(2ap, ap^2)$ is p . Using point-gradient formula,

$$\begin{aligned} \frac{y - ap^2}{x - 2ap} &= p \\ y - ap^2 &= px - 2ap^2 \\ +ap^2 & \quad +ap^2 \\ y &= px - ap^2 \end{aligned}$$

ii. (3 marks) Similarly, equation of tangent at Q is $y = qx - aq^2$. Point of intersection:

$$\begin{aligned} \begin{cases} y = px - ap^2 \\ y = qx - aq^2 \end{cases} \\ px - ap^2 &= qx - aq^2 \\ px - qx &= ap^2 - aq^2 \\ x(p - q) &= a(p - q)(p + q) \\ x &= a(p + q) \end{aligned}$$

Finding y coordinate,

$$\begin{aligned} y &= p(a)(p + q) - ap^2 \\ &= ap^2 + apq - ap^2 = apq \\ \therefore T(a(p + q), apq) \end{aligned}$$

iii. (1 mark)

As T lies on $y = -2a$,

$$\begin{aligned} apq &= -2a \\ \therefore pq &= -2 \end{aligned}$$

Question 6 (Lam)

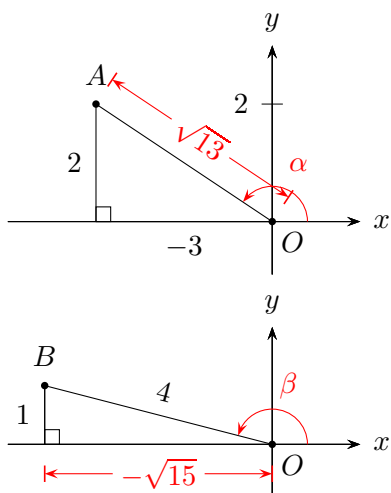
(a) i. (2 marks)

$$\begin{aligned} \frac{\sin 3x}{\cos 2x \cos x} &= \frac{\sin(2x + x)}{\cos 2x \cos x} \\ &= \frac{\sin 2x \cos x + \cos 2x \sin x}{\cos 2x \cos x} \\ &= \frac{\sin 2x \cancel{\cos x} + \cos 2x \sin x}{\cos 2x \cancel{\cos x} \cos x} \\ &= \tan 2x + \tan x \end{aligned}$$

ii. (2 marks)

$$\begin{aligned} \tan 2x + \tan x &= \frac{\sin 3x}{\cos 2x \cos x} = 0 \\ \therefore \sin 3x &= 0 \quad (0 \leq 3x \leq 6\pi) \\ 3x &= 0, \pi, 2\pi, 3\pi, 4\pi, 5\pi, 6\pi \\ x &= 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \frac{5\pi}{3}, 2\pi \end{aligned}$$

(b) $\tan \alpha = -\frac{2}{3}$, $\sin \beta = \frac{1}{4}$ implies second quadrant:



i. (2 marks)

$$\begin{aligned}\sin 2\alpha &= 2 \sin \alpha \cos \alpha \\ &= 2 \times \frac{2}{\sqrt{13}} \times \left(-\frac{3}{\sqrt{13}}\right) \\ &= -\frac{12}{13}\end{aligned}$$

ii. (3 marks)

$$\begin{aligned}\cos(\beta - \alpha) &= \cos \beta \cos \alpha + \sin \beta \sin \alpha \\ &= \left(-\frac{\sqrt{15}}{4}\right) \times \left(-\frac{3}{\sqrt{13}}\right) \\ &\quad + \left(\frac{1}{4}\right) \times \left(\frac{2}{\sqrt{13}}\right) \\ &= \frac{3\sqrt{15} + 2}{4\sqrt{13}}\end{aligned}$$

(c) (3 marks)

$$\begin{cases} y = mx \\ y = 2mx \\ \tan \theta = \frac{1}{3} \end{cases}$$

The angle between two lines:

$$\begin{aligned}\tan \theta &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \left| \frac{2m - m}{1 + 2m^2} \right| \\ &= \left| \frac{m}{1 + 2m^2} \right| = \frac{1}{3}\end{aligned}$$

Case 1:

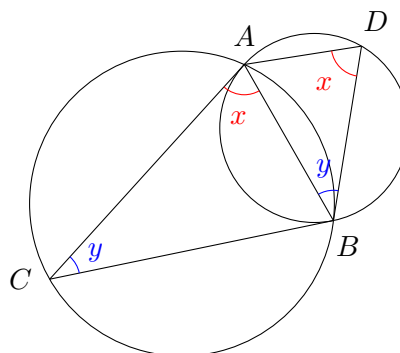
$$\begin{aligned}\frac{m}{1 + 2m^2} &= \frac{1}{3} \\ 3m &= 1 + 2m^2 \\ 2m^2 - 3m + 1 &= 0 \\ (2m - 1)(m - 1) &= 0 \\ \therefore m &= \frac{1}{2}, 1\end{aligned}$$

Case 2:

$$\begin{aligned}\frac{m}{1 + 2m^2} &= -\frac{1}{3} \\ -3m &= 1 + 2m^2 \\ 2m^2 + 3m + 1 &= 0 \\ (2m + 1)(m + 1) &= 0 \\ \therefore m &= -\frac{1}{2}, -1\end{aligned}$$

As $m > 0$, $m = \frac{1}{2}$ or 1 only.

Question 7 (Lin) (3 marks)



- Let $\angle CAB = x$. Then $\angle ADB = x$ ($\angle$ in the alternate segment)
- Let $\angle ABD = y$. Then $\angle ACB = y$ ($\angle$ in the alternate segment)
- In $\triangle ABC$

$$\angle ABC = 180^\circ - (x + y)$$

($\angle$ sum of $\triangle$)

- Similarly in $\triangle ABD$,

$$\angle BAD = 180^\circ - (x + y)$$

- $\therefore AD \parallel BC$:

$$\angle BAD = \angle ABC = 180^\circ - (x + y)$$

and $\angle BAD$ and $\angle ABC$ are alternate angles.

Hence $\angle CBD = \angle CDB$, and

$$2\angle CBD + 60^\circ = 180^\circ$$

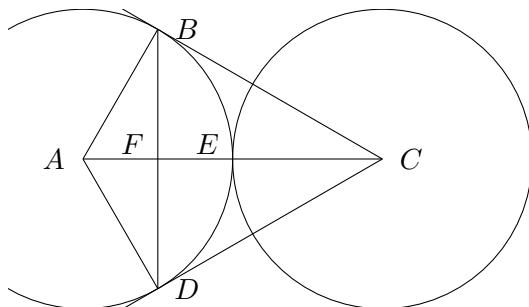
$$2\angle CBD = 120^\circ$$

$$\therefore \angle CBD = 60^\circ = \angle CDB$$

Hence $\triangle CBD$ is equilateral.

Question 8 (Fletcher)

- (a) (2 marks)



- $\angle ADC = 90^\circ$ (radius is $\perp$ to tangent at point of contact)
- Similarly, $\angle ABC = 90^\circ$.
- Hence $ADCB$ is a cyclic quadrilateral, as $\angle ABC + \angle ADC = 180^\circ$ – one pair of opposite $\angle$ supplementary.

- (b) (2 marks)

- $AE = CE$ (given, as both circle have same radii).
- As $\angle ABC = \angle ADC = 90^\circ$, then AC must be the diameter ($\angle$ in a semicircle)

- (c) (2 marks)

- Let $AD = x$. Then $AC = 2x$.
- In $\triangle DCA$, $\sin \angle DCA = \frac{x}{2x} = \frac{1}{2}$
- $\therefore \angle DCA = 30^\circ$. Similarly for $\angle BCA$.

- (d) (2 marks)

In $\triangle CBF$ and $\triangle CDF$,

- FC (common)
- $\angle BCF = \angle DCF = 30^\circ$ (previously proven)
- $DC = BC$ (tangents drawn from external point are equal)

$\therefore \triangle CBF \equiv \triangle CDF$ (SAS).

Question 9 (Lucas)

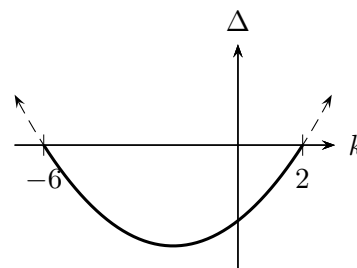
- (a) (2 marks) $a > 0$, $\Delta < 0$ for positive definite.

$$(k+2)^2 - 4(1)(4) < 0$$

$$(k+2)^2 - 16 < 0$$

$$(k+2-4)(k+2+4) < 0$$

$$(k-2)(k+6) < 0$$



$$\therefore -6 < k < 2$$

- (b) i. (2 marks) Let $ax^2 + bx + c = 0$ have roots $x_1 = \tan \alpha$ and $x_2 = \tan \beta$

$$\begin{aligned} \tan(\alpha + \beta) &= \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} \\ &= \frac{x_1 + x_2}{1 - x_1 x_2} \end{aligned}$$

Sum of roots:

$$x_1 + x_2 = -\frac{b}{a}$$

Product of roots:

$$\begin{aligned} x_1 x_2 &= \frac{c}{a} \\ \therefore \tan(\alpha + \beta) &= \frac{-\frac{b}{a}}{1 - \frac{c}{a}} = \frac{-\frac{b}{a}}{\frac{a-c}{a}} \\ &= -\frac{b}{a-c} = \frac{b}{c-a} \end{aligned}$$

ii. (3 marks)

$$\begin{aligned}
& \tan^2(\alpha - \beta) \\
&= \left(\frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta} \right)^2 \\
&= \frac{\tan^2 \alpha - 2 \tan \alpha \tan \beta + \tan^2 \beta}{1 + 2 \tan \alpha \tan \beta + \tan^2 \alpha \tan^2 \beta} \\
&= \frac{x_1^2 + x_2^2 - 2x_1x_2}{1 + 2x_1x_2 + (x_1x_2)^2} \\
&= \frac{(x_1 + x_2)^2 - 2x_1x_2 - 2x_1x_2}{1 + 2x_1x_2 + (x_1x_2)^2} \\
&= \frac{(x_1 + x_2)^2 - 4x_1x_2}{1 + 2x_1x_2 + (x_1x_2)^2} \\
&= \frac{\left(-\frac{b}{a}\right)^2 - \frac{4c}{a}}{1 + \frac{2c}{a} + \frac{c^2}{a^2}} \times \frac{a^2}{a^2} \\
&= \frac{b^2 - 4ac}{a^2 + 2ac + c^2} \\
&= \frac{b^2 - 4ac}{(a + c)^2}
\end{aligned}$$

Question 10 (Ziaziaris)(a) (2 marks) Equation of chord PQ :

$$\begin{aligned}
\frac{y - ap^2}{x - 2ap} &= \frac{ap^2 - aq^2}{2ap - 2aq} \\
\frac{y - ap^2}{x - 2ap} &= \frac{\cancel{a}(p - q)(p + q)}{2\cancel{a}(p - q)} \\
y - ap^2 &= \frac{p + q}{2}x - 2ap \times \frac{p + q}{2} \\
y - ap^2 &= \frac{p + q}{2}x - ap^2 - apq \\
y &= \frac{p + q}{2}x - apq
\end{aligned}$$

As the straight line $y = mx + b$ is also the chord PQ , equate y intercepts:

$$\begin{aligned}
& \therefore -apq = b \\
& \therefore pq = -\frac{b}{a} \quad (10.1)
\end{aligned}$$

(b) (2 marks)

Previously, the y intercepts were equated.
Equate gradients:

$$\begin{aligned}
m &= \frac{p + q}{2} \\
2m &= p + q \quad (10.2) \\
\therefore p^2 + q^2 &= (p + q)^2 - 2pq \\
&= (2m)^2 - 2\left(-\frac{b}{a}\right) \\
&= 4m^2 + \frac{2b}{a} \\
\therefore p^2 + q^2 &= 4m^2 + \frac{2b}{a} \quad (10.3)
\end{aligned}$$

(c) (2 marks)

$$\begin{cases} x = -apq(p + q) & (1) \\ y = a(2 + p^2 + pq + q^2) & (2) \end{cases}$$

(1): substitute from (10.2) and (10.3)

$$x = -a\left(-\frac{b}{a}\right) \times 2m = 2bm$$

(2): substitute from (10.1) and (10.3)

$$\begin{aligned}
y &= a\left(2 + 4m^2 + \frac{2b}{a} + -\frac{b}{a}\right) \\
&= a\left(2 + 4m^2 + \frac{b}{a}\right) \\
&= 2a + 4am^2 + b \\
\therefore N(2bm, 4am^2 + 2a + b)
\end{aligned}$$

(d) (2 marks) - m fixed (constant).

$$\begin{cases} x = 2bm & (3) \\ y = 4am^2 + 2a + b & (4) \end{cases}$$

Change subject of (3) to b : $b = \frac{x}{2m}$, and substitute to (4):

$$\begin{aligned}
y &= 4am^2 + 2a + \frac{x}{2m}2m = 8am^3 + 4am + x \\
x + (-2m)y &= -8am^3 - 4am \\
x + (-2m)y &= a(-2m)^3 + 2a(-2m)
\end{aligned}$$

Which resembles the form

$$x + py = ap^3 + 2ap$$

Hence the locus is a normal to the parabola where $p = -2m$.